

Psoriasis

By: Mamo Feyissa (MSc.)

Introduction

- Psoriasis is a chronic T-lymphocyte–mediated systemic inflammatory disease characterized by recurrent exacerbations and remissions of thickened, erythematous, and scaling plaques and multiple comorbidities, including psoriatic arthritis.

- Genetic predisposition coupled with an unknown precipitating factor triggers an abnormal **T-lymphocytes** mediated immune response .
- This result in **keratinocyte proliferation** and the initial psoriatic skin lesions.
- **Precipitating factors**
 - skin injury**, infection, drugs, smoking, alcohol consumption, obesity, and psychogenic stress.

Genetic susceptibility – on various chromosomes

- The psoriasis susceptibility locus 1 (PSORS1) on chromosome **6p** is a key gene locus, accounting for up to 50% of disease heritability.
- The **major histocompatibility complex** antigen **HLA-Cw6** and **tumor necrosis factor (TNF)- α** are major psoriasis susceptibility genes, along with interleukin (IL)-23 and many other loci.
- There appears to be a general role for T lymphocytes and a specific role for TH17 lymphocytes in psoriasis pathogenesis and as indicators of psoriasis risk.
- Interactions between dermal dendritic cells and activated Th-1 and Th-17 cells in concert with numerous growth factors and **cytokines** (eg, TNF- α , interferon gamma, IL-1) cause epidermal hyperplasia and dermal inflammation.

- **Plaque psoriasis (psoriasis vulgaris)** is seen in ~90% of psoriasis pts.
 - Lesions are erythematous, red-violet in color, at least 0.5 cm in diameter, well demarcated, and typically covered with silver flaking scales.
 - They may appear as **single lesions** at predisposed areas (eg, knees and elbows) or generalized over a wide body surface area (BSA).
- **Pruritus** may be severe and require treatment to minimize excoriations from frequent scratching.
 - Lesions may be physically debilitating or socially isolating.

Clinical Presentation....

- Preexisting psoriasis can be exacerbated by
 - drugs lithium, NSAIDs, antimalarials, β -adrenergic blockers, fluoxetine, and withdrawal of corticosteroids),
 - times of stress, and
 - seasonal changes.
- Potential comorbidities** include
 - Psoriatic arthritis, depression, anxiety, hypertension, obesity, diabetes mellitus, Crohn's disease, and alcoholism
- Psoriatic arthritis** develops in about 30% of pts with plaque psoriasis.
 - mostly as **polyarticular peripheral arthritis** but can vary widely with peripheral and/or axial, monoarticular, or polyarticular patterns.



A



B

Figure 1. A and B, Patient with limited disease (**<5% BSA**). There are erythematous, predominantly discoid plaques with overlying silvery scale involving the elbows, knees, periumbilical area, back, and buttocks.



Fig 2. Patient with generalized disease treated with UVB. A, 15% BSA showing erythematous, large plaques with silvery scales on the trunk and upper extremities B, Mild palmar involvement.



Fig 3: Recalcitrant psoriasis of the plantar surfaces



Fig 4. Large confluent plaques involving the scalp (A) and temple (B). After three days of topical clobetasol solution twice daily, there was significant improvement. Thick, indurated large scaly plaques are present in the suprapubic area (C).

DIAGNOSIS

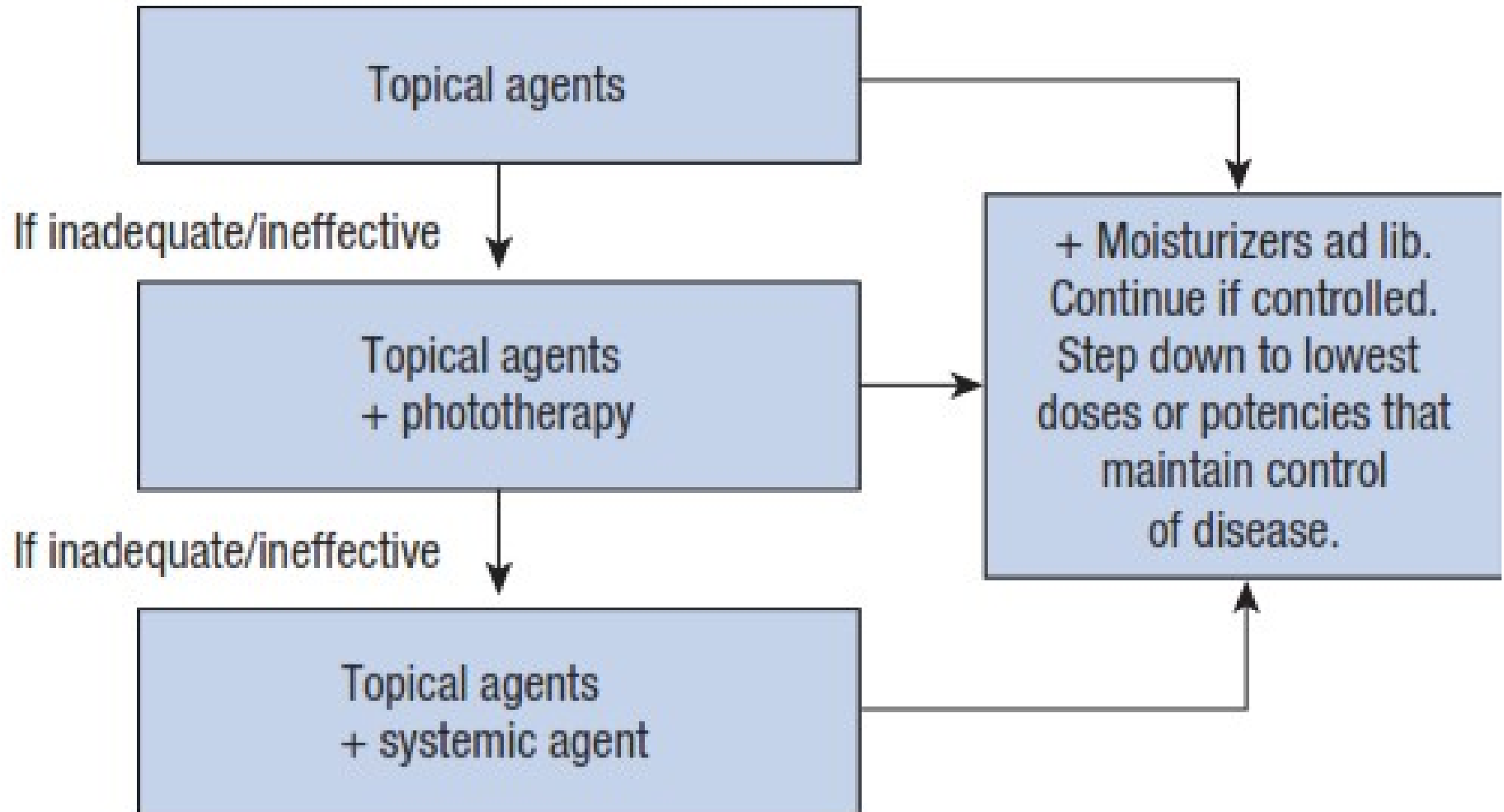
- Diagnosis is based on **physical examination findings of characteristic lesions.**
- Skin biopsies are not diagnostic of psoriasis.
- Classification of psoriasis as **mild**, **moderate**, or **severe** is based on BSA and Psoriasis Area and Severity Index (**PASI**) measurements.
- A 2011 European classification system defines severity of plaque psoriasis as either mild or moderate-to-severe.

Goals of Treatment

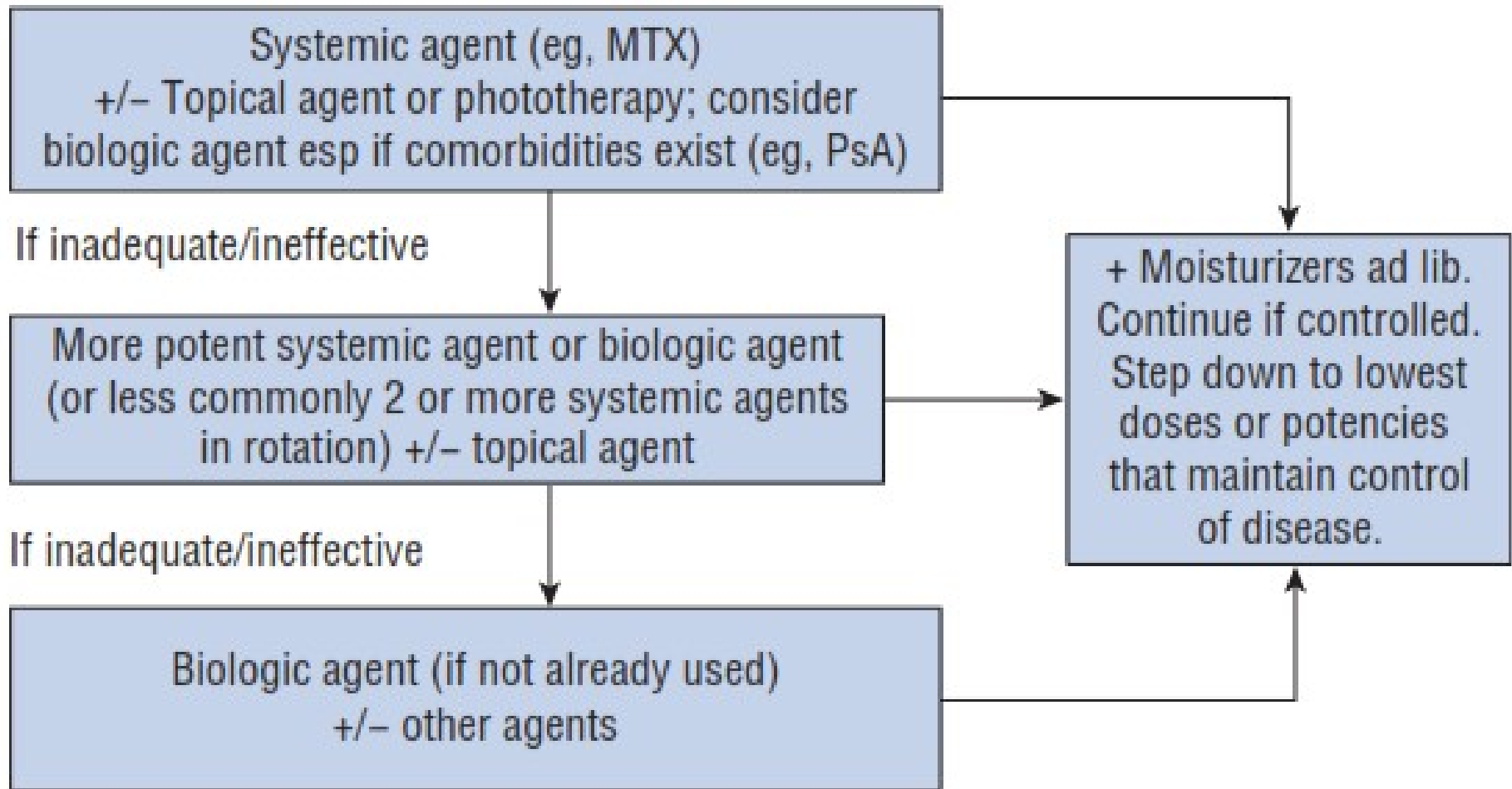
- Minimize or eliminate skin lesions,
- alleviate pruritus,
- reduce frequency of flare-ups,
- treat comorbid conditions,
- screen for and manage lifestyle factors that may trigger exacerbations
- avoid adverse treatment effects,
- provide cost-effective treatment,
- provide appropriate counseling (eg, stress reduction), and
- maintain or improve quality of life.

- **Nonmedicated moisturizers** help maintain skin moisture, reduce skin shedding, control scaling, and reduce pruritus.
- **Oatmeal baths** further reduce pruritus, and regular use may reduce need for systemic antipruritic drugs. Harsh soaps and detergents should be avoided.
- Cleansing should involve tepid water, preferably with lipid- and fragrance-free cleansers.
- **Sunscreens (SPF of 30 and above)** should be used when outdoors.
- **Stress management** can improve extent and severity of psoriasis.

Treatment algorithm for mild to moderate psoriasis.



Treatment algorithm for moderate to severe psoriasis



MTX, methotrexate; PsA, psoriatic arthritis.

Topical Therapies

- **Corticosteroids** have anti-inflammatory, anti-proliferative, immunosuppressive, and vasoconstrictive effects.
- first-line treatment for limited psoriasis either as monotherapy or with nonsteroidal topical agents
- **Lower-potency** products should be used for infants and for lesions on the face, intertriginous areas, and areas with thin skin.
- **Mid- to high-potency** agents are recommended as initial therapy for other areas of the body in adults.
- **Reserve the highest potency** corticosteroids for pts with very thick plaques or recalcitrant dx, such as plaques on the palms and soles.
 - Use potency class I corticosteroids for **only 2 to 4 weeks**.

- **Calcipotriene** is a synthetic vitamin D3 analogue that binds to vitamin D receptors, which inhibits keratinocyte proliferation and enhances keratinocyte differentiation.
 - Vitamin D3 analogues also inhibit **T-lymphocyte activity**
 - Calcipotriene 0.005% cream, ointment, or solution is applied one or two times daily
 - For **mild psoriasis**, calcipotriene is more effective than anthralin and comparable to or slightly more effective than class 3 (upper mid-strength) topical corticosteroid ointments.

- **Tazarotene** is a topical retinoid that normalizes keratinocyte differentiation, diminishes keratinocyte hyperproliferation, and clears the inflammatory infiltrate in psoriatic plaques.
 - As a 0.05% or 0.1% gel and cream applied once daily (evening).
- **Anthralin** has a direct anti-proliferative effect on epidermal keratinocytes, normalizing keratinocyte differentiation.
 - Short-contact anthralin therapy (SCAT) is the preferred regimen, with **ointment applied** only to the thick plaque lesions for 2 hours or less and then **wiped off**.

- **Coal tar** is keratolytic and may have anti-proliferative and anti-inflammatory effects.
 - Formulations include crude coal tar and tar distillates (liquor carbonis detergens) in ointments, creams, and shampoos.
 - **Coal tar** is used infrequently due to limited efficacy and poor pt adherence and acceptance. It has a slower onset of action than calcipotriene, has an unpleasant odor, and stains clothing.
- **Salicylic acid** has keratolytic properties and has been used in shampoos or bath oils for scalp psoriasis.
 - It enhances penetration of topical corticosteroids, thereby increasing corticosteroid efficacy.

- **Pimecrolimus** 1% cream is a calcineurin inhibitor shown to be effective for plaque psoriasis when used under occlusion and for pts with **moderate-to-severe** inverse psoriasis (involving intertriginous areas).
- It may be a useful alternative for pts with intertriginous or facial lesions because it is
 - less irritating than calcipotriene and
 - does not have the topical adverse effects of corticosteroids (eg, skin atrophy).

- Phototherapy consists of nonionizing electromagnetic radiation, either ultraviolet A (UVA) or ultraviolet B (UVB), as light therapy for psoriatic lesions.
- Broadband UVB is also given as photochemotherapy with topical agents such as crude coal tar or anthralin for enhanced efficacy.
- Adverse effects include erythema, pruritus, xerosis, hyperpigmentation, and blistering.

- **Acitretin** is a retinoic acid derivative and the active metabolite of etretinate.
- Retinoids may be **less effective** than **methotrexate** or cyclosporine when used as monotherapy.
- Acitretin is more commonly used in combination with topical calcipotriene or phototherapy. Initial recommended dose is 25/50 mg;
- It is better tolerated when taken with meals
- All retinoids are teratogenic and are pregnancy category X. Acitretin should not be used in women of childbearing potential.

- **Cyclosporine** is a systemic calcineurin inhibitor that is effective for inducing remission and for maintenance therapy of moderate to severe plaque psoriasis.
- It is also effective for pustular, erythrodermic, and nail psoriasis.
- Intermittent short-course therapy (<12 weeks) is preferable to continuous therapy because it appears to reduce the risk of nephrotoxicity.
 - Cyclosporine is significantly more effective than etretinate and has similar or slightly better efficacy than methotrexate.
 - The usual dose is between **2.5 and 5 mg/kg/day bid**. Maintenance therapy using low doses (1.25–3 mg/ kg/day) may prevent relapse.

- **Methotrexate** has antiinflammatory effects due to its effects on T-cell gene expression and also has cytostatic effects.
- It is more effective than acitretin and has similar or slightly less efficacy than cyclosporine.
- Methotrexate can be administered orally, subcutaneously, or intramuscularly.
- The starting dose is 7.5 to 15 mg once weekly, increased incrementally by 2.5 mg every 2 to 4 weeks until response; maximal doses are 25 mg weekly.

- **Biologic response modifiers (BRMs)** are considered for moderate to severe psoriasis when other systemic agents are inadequate or contraindicated.
- Cost considerations tend to limit their use as first-line therapy
- BRMs include:
 - Adalimumab
 - Etanercept
 - Infliximab
 - Alefacept
 - Ustekinumab

- Combination therapy may be used to enhance efficacy or minimize toxicity.
- Combinations can include two topical agents, a topical agent plus phototherapy, a systemic agent plus topical therapy, a systemic agent plus phototherapy, two systemic agents used in rotation, or a systemic agent and a BRM

- **Mycophenolate mofetil** inhibits DNA and RNA synthesis and may have a lymphocyte anti-proliferative effect.
 - Oral mycophenolate mofetil may be effective in some cases of moderate to severe plaque psoriasis.
 - The usual dose is 500 mg orally four times daily, up to a max. of 4 g daily.
- **Hydroxyurea** inhibits cell synthesis in the S phase of the DNA cycle.
 - It is sometimes used for pts with recalcitrant severe psoriasis, but BRMs may be a better option in these pts. The typical dose is 1 g daily, with a gradual increase to 2 g daily as needed and as tolerated.

EVALUATION OF THERAPEUTIC OUTCOMES

- Help patients understand the general principles of therapy and the importance of adherence.
- A positive response involves normalization of involved areas of skin, as measured by reduced erythema and scaling, as well as reduction of plaque elevation improved quality of life.
- Achievement of efficacy by any therapeutic regimen requires days to weeks.
- Initial dramatic response may be achieved with some agents, such as corticosteroids.